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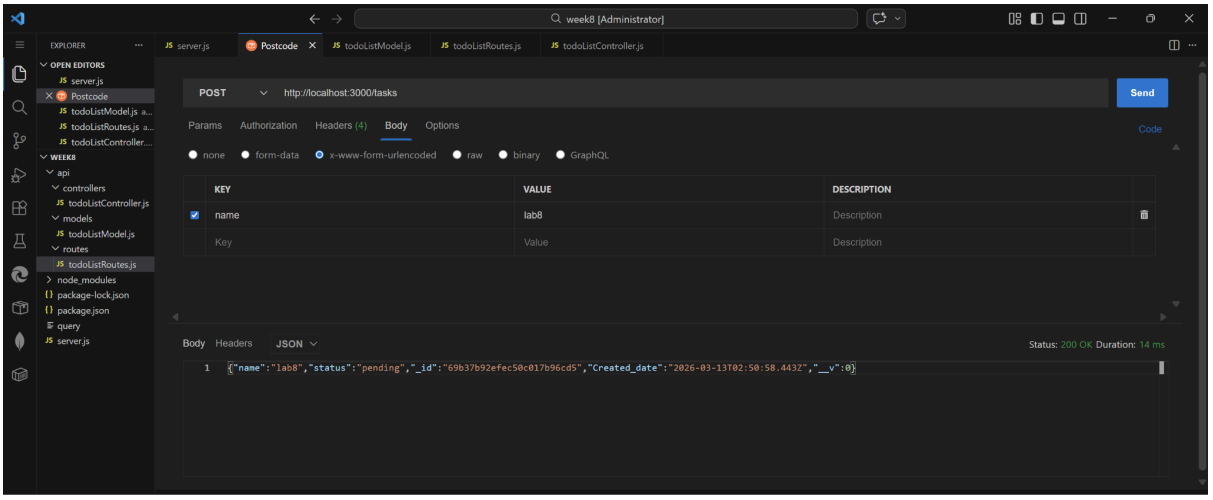
RESTful API

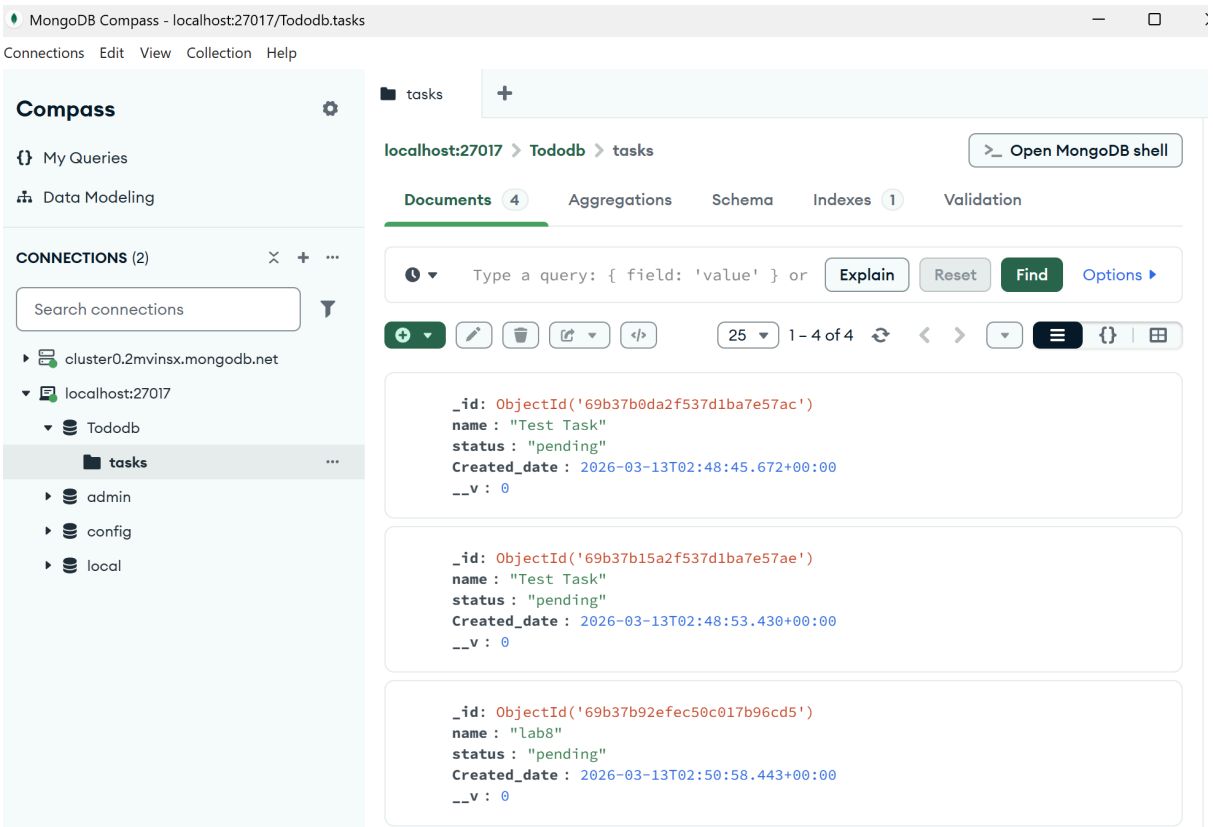
Image

```
const TaskSchema = new Schema({
    name:{
        type: String,
        required: 'Enter a name for the task'
    },
    Created_date:{
        type: Date,
        default: Date.now
    },
    status:{
        type:[{
        type: String,
        enum: ['pending','ongoing','completed']
    }],
    default: ['pending']
    }
});

module.exports = mongoose.model('Task', TaskSchema);
```

```
PS C:\Users\Admin\Desktop\Study\COMP-1842\week8> nodemon server.js
[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
todo list RESTful API server running on port 3000
```





Understanding

Through this week's lectures and hands-on lab tasks, I have developed a clear understanding of RESTful APIs and their crucial role in web development. I now understand that an API acts as a communication bridge between the client and the server, and REST provides a standardized, stateless approach to manage this interaction. I learned how to map CRUD operations directly to HTTP methods: GET for retrieving data, POST for creating, PUT for updating, and DELETE for removing resources.

During the lab task, I successfully built a Todo List API from scratch. This practical experience solidified my understanding of backend architecture, specifically how to organize code into Models (defining the data schema with Mongoose), Controllers (writing the core

logic), and Routes (handling URL endpoints). Furthermore, applying these concepts to our Vocab Builder coursework showed me how a robust Node.js backend prepares data to be consumed by a frontend framework like Vue.js.

One of the most valuable skills I acquired was testing. Using the Postcode extension, I learned how to simulate client requests, send JSON data using body-parser, and verify the server's responses. Seeing the data successfully appear in my MongoDB database was incredibly rewarding and proved that my entire backend system was functioning correctly.